

2.CANDIDATE ELIMINATION ALGORITHM

```
import pandas as pd
# Read CSV file
df = pd.read_csv("find s - Sheet1 (1).csv")
# Rename the last column
df.rename(columns={df.columns[-1]: "Allergy"}, inplace=True)
print("Training Data\n")
print(df)
# Ignore Number column
concepts = df.iloc[:, 1:-1].values
target = df.iloc[:, -1].values
# Initialize S and G
S = ['Ø'] * len(concepts[0])
G = [['?'] * len(concepts[0])]
print("\nInitial Specific Boundary (S):", S)
print("Initial General Boundary (G):", G)
for i, h in enumerate(concepts):

    if target[i] == "Yes":

        if S == ['Ø'] * len(concepts[0]):
            S = list(h)
        else:
            for j in range(len(S)):
                if S[j] != h[j]:
                    S[j] = '?'

        G = [g for g in G if all(g[k] == '?' or g[k] == h[k] for k in range(len(g)))]

    else:

        new_G = []

        for g in G:
            for j in range(len(g)):
                if g[j] == '?':
                    if S[j] != '?' and S[j] != h[j]:
                        new_hypothesis = g.copy()
                        new_hypothesis[j] = S[j]
```

```
if new_hypothesis not in new_G:  
    new_G.append(new_hypothesis)
```

```
G = new_G
```

```
print("\nAfter Example", i + 1)
```

```
print("S =", S)
```

```
print("G =", G)
```

```
print("\nFinal Specific Boundary (S):", S)
```

```
print("Final General Boundary (G):", G)
```

OUTPUT

Initial Specific Boundary (S): ['Ø', 'Ø', 'Ø', 'Ø']
Initial General Boundary (G): [['?', '?', '?', '?']]

Example 1

Instance : ["Sam's" 'Breakfast' 'Friday' 'Cheap']
Target : Yes
Specific Boundary (S): ["Sam's", 'Breakfast', 'Friday', 'Cheap']
General Boundary (G): [['?', '?', '?', '?']]

Example 2

Instance : ["Kim's" 'Lunch' 'Friday' 'Expensive']
Target : No
Specific Boundary (S): ["Sam's", 'Breakfast', 'Friday', 'Cheap']
General Boundary (G): [["Sam's", '?', '?', '?'], ['?', 'Breakfast', '?', '?'], ['?', '?', '?', 'Cheap']]

Example 3

Instance : ["Sam's" 'Lunch' 'Saturday' 'Cheap']
Target : Yes
Specific Boundary (S): ["Sam's", '?', '?', 'Cheap']
General Boundary (G): [["Sam's", '?', '?', '?'], ['?', '?', '?', 'Cheap']]

Example 4

Instance : ["Bob's" 'Breakfast' 'Sunday' 'Cheap']
Target : No
Specific Boundary (S): ["Sam's", '?', '?', 'Cheap']
General Boundary (G): [["Sam's", '?', '?', '?'], ["Sam's", '?', '?', 'Cheap']]

Example 5

Instance : ["Sam's" 'Breakfast' 'Sunday' 'Expensive']
Target : No
Specific Boundary (S): ["Sam's", '?', '?', 'Cheap']
General Boundary (G): [["Sam's", '?', '?', 'Cheap'], ["Sam's", '?', '?', 'Cheap']]

=====
Final Specific Boundary (S): ["Sam's", '?', '?', 'Cheap']
Final General Boundary (G): [["Sam's", '?', '?', 'Cheap'], ["Sam's", '?', '?', 'Cheap']]